



Research Proposal

**Treatment of various degrees of white spot lesions using resin infiltration—in vitro
study**

Student Name:

Level:

Department:

Co-supervisor:

Affiliation:

Introduction

The first clinical sign of enamel demineralisation is a loss of minerals under the intact enamel surface, which presents as opaque white areas on the surface of the teeth and is referred to as white spot lesions (WSLs). Accessibility for cleaning while undergoing fixed orthodontics is one of the most frequent problems. Bacterial biofilms can also develop around orthodontic appliances and may increase acid production, causing an increase in demineralization of the enamel. The WSLs can be seen within a few weeks following treatment and can last for a while after the appliances are removed. In addition to being a risk factor for dental caries, these lesions have an impact on the aesthetics of the patient's smile, and therefore the patient's satisfaction, especially for anterior teeth (Patano et al., 2023; Bourouni et al., 2021).

The organic acids produced by the cariogenic bacteria begin to break down the calcium and phosphate ions from the surface of the enamel, and WSLs begin to develop. The outer enamel is generally rather well preserved, but there is some porosity within the enamel, resulting in light scattering, and hence the characteristic chalky white appearance. The severity and visibility of WSLs depend on the depth of the lesions and the extent of the mineralization loss within the lesions when compared to the mineralization of the sound enamel. This means that deeper lesions will pose a greater challenge to masking and will not be as likely to be treated conservatively. Early diagnosis of the lesion is thus crucial to prevent propagation of the lesion, and early intervention is important before the onset of cavitation.

There are a number of proposed management strategies for WSLs. Fluoride treatments have also been found to be popular since they strengthen enamel by remineralizing and also render it less vulnerable to acid. Similarly, minerals replenishing agents (e.g., casein phosphopeptide-amorphous calcium phosphate (CPP-ACP)) have been proven to replenish lost minerals and retard lesion progression. There are other techniques, such as enamel microabrasion, bleaching, and restoration using composite resin restorations. But not all of these have any disadvantages. Other therapies, such as fluoride and remineralization therapy, will improve/mineralize the lesion, but will not be enough to eliminate the esthetic appearance of lesions. Other therapies, such as microabrasion and restorative therapy, will cause irreversible removal of enamel and may be more invasive (Patano et al., 2023; Marya et al., 2022).

However, all these disadvantages can be overcome by employing minimally invasive procedures for the treatment of non-cavitated WSLs, such as recently developed resin infiltration. It is a technique in which the outer enamel is hypermineralized and removed from the enamel surface, while the lesion body is porous and is infiltrated by the resin with low viscosity. The infiltrant is able to penetrate into the microporosities of the enamel and alters the refractive index of the lesion, which decreases light scattering and improves the appearance. The most popular system is the ICON resin infiltrant system that integrates hydrochloric acid etching, ethanol drying, and resin infiltration. This will not only catch the lesion, but will also provide immediate aesthetic improvement with maintenance of healthy tooth structure (Bourouni et al., 2021).

Need for the Study

Although resin infiltration of orthodontically induced WSLs has been successful in some recent studies. Resin infiltration is effective in masking lesions in systematic reviews when compared to natural remineralisation and fluoride treatments. Other findings showed that the effectiveness of treatment is related to the depth of the lesion, and there is better aesthetic improvement in superficial enamel lesions as compared with deeper dentinal lesions. Although there is increasing evidence for the effectiveness of resin infiltration, there is limited research about the optimal sequence of application of both etching and infiltrant for lesions of different depths. Major studies tend to be on the overall effectiveness of treatment and not on optimizing the protocol. Identifying the most appropriate treatment plan can provide better clinical response, prevent unnecessary erosion of enamel, and help the clinician make evidence-based recommendations for management of varying amounts of white spot lesions.

Expected Benefits of the Study

The results of this study will provide evidence-based clarity of the differences in resin infiltration depending on the depth of white spot lesions and the sequence of application. The results can help clinicians choose the best infiltration protocol to achieve an optimal aesthetic outcome without compromising the structure of the enamel. Besides, the results may be used to optimize the use of minimally invasive therapies for orthodontically induced white spot lesions, particularly when simple remineralisation therapies fail. Lastly, the study may be useful for predicting the efficiency of lesions masking and the standardisation of the protocols of different types of lesions.

Aim of the Study

This study aims to evaluate the effectiveness of different resin infiltration methods in the color and appearance of different depths of white spot lesions (WSLs). The second objective of the study is to understand the best sequence of the different etching and infiltration with resins on each type of lesion that will result in the highest enhancement in esthetics without removing unnecessary enamel.

Null Hypothesis

The colour improvement will not be significantly different for White spot lesions of various depths after the various resin infiltration treatments.

Research Question

Are there any differences in the effectiveness of resin infiltration with lesion depth and the number of applications of etching and infiltrant?

Materials and Methodology

Ethical Approval

IREC ethical clearance will be sought prior to the study starting. All the activities of collection, storage, handling, and disposal of biological samples for research will be conducted according to the accepted ethical and biosafety procedures, as all research will be conducted using teeth extracted from humans. Teeth will only be collected after they have been extracted for orthodontic reasons, and they will be anonymized to ensure that they are kept in a confidential manner, in compliance with ethical standards (Patano et al., 2023). All extracted teeth will be acquired from patients who have given informed consent for the use of extracted biological material for research use, as per institutional and clinical dental extraction procedures. Ethical approval documentation, including the approved letter and the submission forms (Institutional Research Ethics Committee (IREC) approval letter), will be collected before the study and submitted with the final research record.

Study Design

It is an experimental comparative in vitro study and will compare various resin infiltration methods in terms of colour and appearance for various depths of white spot lesions (WSLs). An in vitro design can be useful to manage experimental conditions to assess the impact of treatments, without confounding factors associated with patients.

Study Samples and Selection

Inclusion Criteria

The study sample will consist of extracted human premolars meeting the following criteria:

- Sound teeth free from caries.
- Intact buccal enamel surfaces.
- No visible cracks or fractures.
- No enamel hypoplasia.
- No previous restorations or chemical treatments.

Exclusion Criteria

The following teeth will be excluded from the study:

- Teeth affected by dental caries.
- Teeth exhibiting enamel defects or hypoplasia.
- Teeth with restorations.
- Teeth presenting visible cracks, fractures, or structural damage.

Sample Size Calculation

The study will be based on 96 human premolars that will be extracted. The lesions will be classified into four categories of teeth, with 24 teeth in each category, according to the depth of lesions. These will then be split into a set of four treatment subgroups with six teeth in each of the four. Sample size is similar to other experimental research about the success of resin infiltration, and they are enough to reveal the significance of the difference between treatments (Abbas et al., 2018).

Sample Preparation

All extracted teeth will be completely cleaned with artificial saliva, and the conditions of the mouth will be simulated. The roots will be taken off at the cemento-enamel junction with a diamond disk. The teeth will then be placed in a self-curing acrylic resin for colour evaluation, thus maintaining the same positions. Baseline colour measurements (T0) will be measured using a calibrated spectrophotometer (measured in the Commission Internationale de l'Eclairage (CIE lab*) colour system). Standardized positioning devices will be used to minimize variability in measurements and to improve the reliability (Tavares et al., 2021).

Induction and Classification of White Spot Lesions

A controlled demineralization process using an acid solution in which the mineral content of the enamel will be simulated will form white spot lesions. After induction of lesions, the depth of demineralization will then be measured using the DIAGNOdent diagnostic system. Based on the DIAGNOdent reading, the teeth will be categorized into 4 different types of lesions:

- Shallow enamel lesions
- Deep enamel lesions
- Shallow dentine lesions
- Deep dentine lesions

Based on this classification, the success of the treatment with different degrees of lesions can be determined.

Intervention Procedures

Following the categorization of lesions, each depth group will be treated with the 4 different resin infiltration treatments of the ICON resin infiltrant system, in a random manner.

Group EA

- One etching application
- One resin infiltrant application

Group EAA

- One etching application
- Two resin infiltrant applications

Group EEA

- Two etching applications
- One resin infiltrant application

Group EEAA

- Two etching applications
- Two resin infiltrant applications

All procedures will be performed according to the manufacturer's instructions, for uniformity and repeatability.

Outcome Assessment

Resin infiltration will be the primary outcome measure. Using a spectrophotometer, color measurements will be taken at three different times: baseline (T0), lesion formation (T1), and treatment (T2). The colour differences will be presented as ΔE , which can be quantitatively assessed to evaluate the aesthetic improvement. The lower the ΔE baseline to post-treatment scores, the more lesions will be masked and the better the colour recovery (Bourouni et al., 2021).

Statistical Analysis

Data analysis will be done using IBM SPSS software. All groups will have descriptive statistics (mean and standard deviation) calculated. The mean difference in the colour of the various treatment protocols will be compared to the mean difference of the various categories of lesion-depth, using One-way Analysis of Variance (ANOVA). When differences are large, then post-hoc multiple comparisons will be done to identify the differences between groups. The statistical significance of the numbers will be considered at $P < 0.05$.

Timeline for Research

Research Phase	Duration
Proposal Writing	Week 1–2
Literature Review	Week 3–4
Sample Collection	Week 5
Sample Preparation	Week 6
White Spot Lesion (WSL) Induction	Week 7–9
Resin Infiltration Procedures	Week 10–11
Data Collection	Week 12
Statistical Analysis	Week 13
Report Writing and Interpretation of Findings	Week 14–15

Research Budget

Item	Estimated Cost (Riyal)
Extracted Teeth Collection	300
Artificial Saliva Preparation	200
Demineralization Chemicals	400
DIAGNOdent Usage	800
Spectrophotometer Usage	1000
ICON Resin Infiltration Kits	1800
Acrylic Resin Materials	250
Statistical Analysis	250
Miscellaneous Laboratory Expenses	500
Total Estimated Budget	5500 Riyal

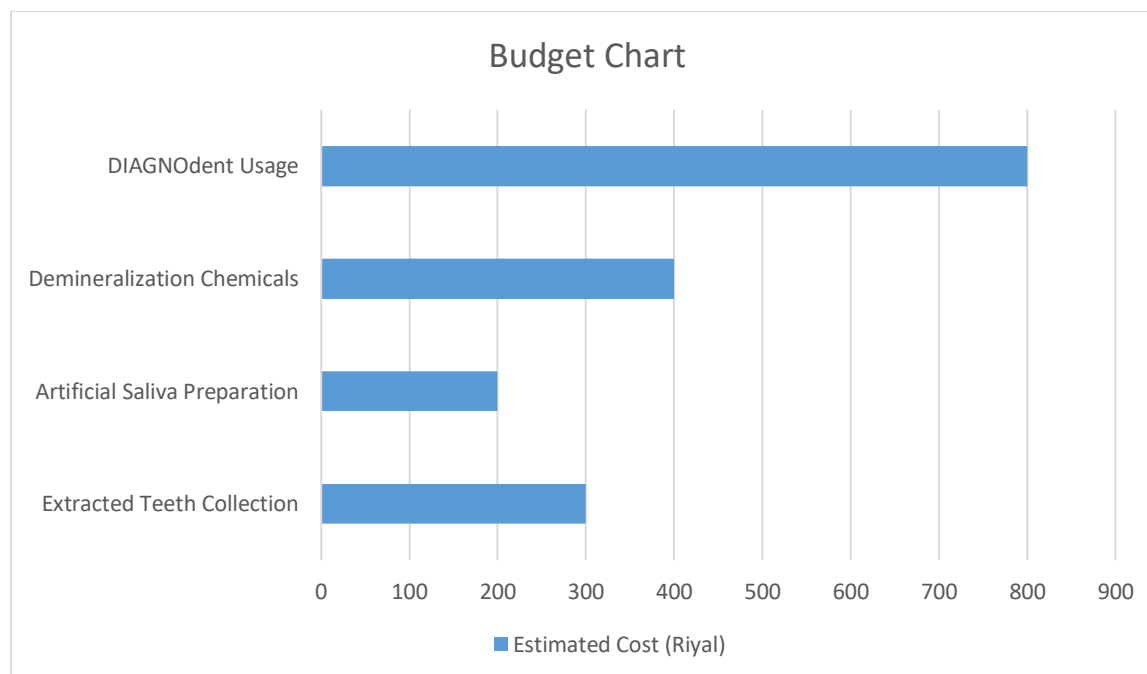


Figure 1: Budget Chart

The above budgets are approximate and may vary depending on facilities, equipment, and institutional support.

Bibliography

- Bourouni, S., Dritsas, K., Kloukos, D. and Wierichs, R.J. (2021). Efficacy of resin infiltration to mask post-orthodontic or non-post-orthodontic white spot lesions or fluorosis — a systematic review and meta-analysis. *Clinical Oral Investigations*, 25(8), pp.4711–4719. doi:<https://doi.org/10.1007/s00784-021-03931-7>.
- Lazar, L., Vlasa, A., Beresescu, L., Bud, A., Lazar, A.P., Matei, L., and Bud, E. (2023). White Spot Lesions (WSLs)-Post-Orthodontic Occurrence, Management and Treatment Alternatives: A Narrative Review. *Journal of Clinical Medicine*, [online] 12(5), p.1908. doi:<https://doi.org/10.3390/jcm12051908>.

- Marya, A., Venugopal, A., Karobari, M.I., and Rokaya, D. (2022). White Spot Lesions: A Serious but Often Ignored Complication of Orthodontic Treatment. *The Open Dentistry Journal*, 16(1). doi:<https://doi.org/10.2174/18742106-v16-e2202230>.
- Patano, A., Malcangi, G., Sardano, R., Mastrodonato, A., Garofoli, G., Mancini, A., Inchingolo, A.D., Di Venere, D., Inchingolo, F., Dipalma, G. and Inchingolo, A.M. (2023). White Spots: Prevention in Orthodontics-Systematic Review of the Literature. *International Journal of Environmental Research and Public Health*, [online] 20(8), p.5608. doi:<https://doi.org/10.3390/ijerph20085608>.
- Tavares, M.I., Saraiva, J., do Vale, F., Coelho, A.S., Amaro, I.F., Marto, C.M., Francisco, I., Ferreira, M.M., Paula, A.B. and Carrilho, E.V. (2021). Resin infiltration in white spot lesions caused by orthodontic hypomineralisation: a minimally invasive therapy. *British Dental Journal*, [online] 231(7), pp.387–392. doi:<https://doi.org/10.1038/s41415-021-3476-z>.

Here are the **partially met sections only**:

Section	Status	What is missing / needs improvement
Researchers' Information	Partly met	The required fields are present, but they are not filled in. Add student name, academic level, department, supervisor name, co-supervisor name, qualifications, and department.
Expected Benefits	Partly met	You mentioned benefits inside the "Need for the Study," but they are not clearly separated. Add a short paragraph titled Expected Benefits of the Study .
Informed Consent / Human Samples	Partly met	You mention extracted human teeth and anonymization, but you should add a clear sentence about consent or permission for using extracted teeth.
Ethics Requirement Evidence	Partly met	You state that IREC ethical clearance will be sought, but you should mention that the ethics form/approval evidence will be attached or submitted.

Appendices

Appendix A: Ethical approval form

Institutional Research Ethics Committee (IREC) Application

- Study Title: Treatment of Various Degrees of White Spot Lesions Using Resin Infiltration
– In Vitro Study
- Principal Investigator: _____
- Supervisor: _____
- Department: _____
- Institution: _____
- Study Design: In vitro experimental study using extracted human premolars
- Sample Source: Extracted orthodontic teeth (anonymised)

Ethical Considerations:

- No direct patient contact involved
- Teeth will be anonymised and used solely for research purposes
- Biological samples will be handled according to biosafety protocols
- No identifiable patient data will be recorded

Declaration:

This study will commence only after formal ethical approval is granted by IREC.

Appendix B: Consent/source permission form for extracted teeth, if required

Consent for Use of Extracted Teeth in Research

I confirm that my extracted tooth/teeth may be used for educational and research purposes. I understand that:

- The tooth will be anonymised immediately after extraction
- No personal identifiers will be attached
- The sample will be used strictly for laboratory-based research

- There is no risk or impact on my treatment outcome

Patient Name: _____

Signature: _____

Date: _____

Clinician: _____

Appendix C: Budget support documents or invoices, if available

Research Budget Justification Sheet

Item	Estimated Cost	Basis
ICON resin infiltration kits	1800 SAR	Supplier quotation
DIAGNOdent usage	800 SAR	Institutional lab rate
Spectrophotometer usage	1000 SAR	Lab hourly charge
Chemicals & demineralization agents	400 SAR	Market estimate
Artificial saliva preparation	200 SAR	Lab supply estimate
Acrylic resin materials	250 SAR	Supplier estimate
Miscellaneous expenses	500 SAR	Contingency
Statistical analysis support	250 SAR	Software/lab support

Note: Final invoices will be attached upon procurement.

Appendix D: Research application form

Institutional Research Application Summary

- Project Title: Treatment of White Spot Lesions Using Resin Infiltration
- Research Type: Experimental in vitro study
- Objective: To evaluate the effect of resin infiltration protocols on different lesion depths
- Methodology Summary: Use of extracted premolars, lesion induction, resin infiltration, spectrophotometric analysis
- Sample Size: 96 extracted human premolars
- Risk Level: Minimal (no human subjects directly involved)

- Data Handling: Non-identifiable laboratory data only
- Storage: Secure laboratory database

Declaration:

All procedures will comply with institutional research and biosafety regulations.